Jute Fibre Straightening Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Jute Fibre Straightening Machine

What it is used for

To straighten and align jute fibres for subsequent processing, spinning, or product manufacture. To achieve uniform fibre alignment while minimizing fibre damage and keeping the machine free from blockages.

Main parts

- Machine frame and feed table
- Feed rollers or fibre guides
- Straightening combs or rollers
- Drive motor and transmission
- Speed controls and emergency stop
- Protective guards and fibre collection area

Before you start

1. Read the operating manual and confirm the machine is suitable for the jute fibre being processed.
2. Inspect feed rollers, combs, guides, guards, belts, and electrical connections for damage.
3. Ensure the machine is securely installed and all protective covers are fitted.
4. Check that the fibre feed area and discharge path are free from previous material and obstructions.
5. Prepare jute fibres by removing stones, metal pieces, strings, and other foreign objects.
6. Wear suitable eye protection, close-fitting clothing, and respiratory protection where airborne dust is present.

How to use

1. Switch off and isolate power before adjusting guides or inspecting moving parts.
2. Set the feed guides and operating speed according to the fibre condition and machine instructions.
3. Confirm that the discharge and fibre collection areas are ready.
4. Start the machine using the approved procedure and observe it for unusual noise or vibration.
5. Feed a small quantity of jute fibres evenly into the designated feed area.
6. Allow the rollers or combing mechanism to straighten the fibres without forcing the material.
7. Increase the feed gradually while checking fibre alignment and machine loading.
8. Monitor the output for knots, uneven alignment, excessive breakage, or blockage.
9. Stop the machine if fibres wrap around rollers or the feed becomes irregular.

10. Isolate power and wait for all moving parts to stop before clearing any blockage.
11. Collect the processed fibres and inspect them for consistency and damage.
12. Switch off the machine and clean fibre residues using the approved method.

⚠ Safety

- Only trained operators should use the machine.
- Never reach into feed rollers, combs, belts, or other moving parts.
- Do not operate the machine with missing guards or damaged safety controls.
- Tie back long hair and avoid loose clothing, jewellery, or dangling accessories.
- Isolate power before removing tangled fibres or performing adjustments and maintenance.
- Use appropriate dust protection and ventilation where fine fibres or dust are generated.

Care & storage

- Remove fibre waste from the feed table, rollers, and collection area after use.
- Inspect combs, rollers, belts, and guards regularly for wear or damage.
- Lubricate only specified points according to the manual.
- Keep electrical components dry and clean ventilation openings.
- Record faults and arrange servicing according to the maintenance schedule.

Common mistakes to avoid

- Feeding excessive fibre at once and overloading the machine.
- Using fibres containing hard foreign objects that can damage rollers or combs.
- Pulling tangled fibres out while the machine is running.
- Operating with damaged guards or worn feed components.
- Ignoring uneven output, excessive vibration, or repeated blockages.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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